SPECTROSCOPIC APPLICATIONS OF THE FABRY–PÉROT INTERFEROMETER

OPTICS LABORATORY REPORT

Determination of Laser Wavelength and
Sodium Doublet Separation

Submitted by

Partha Sarathi Mohanta

Roll No.: 2511111

Experimental Group Members

Piyush Sharma	2511115
Pranav Lalit Chandak	2511118

Code repository: github.com/ParthaXD/Interferrometers

Light Sources:	He–Ne laser and sodium source
Instrument:	Fabry–Pérot interferometer
Measurements:	Laser wavelength and doublet separation

National Institute of Science Education and Research

Bhubaneswar

2026

Abstract

The Fabry–Pérot interferometer is a multiple-beam interference instrument capable of resolving closely spaced spectral features through the formation of sharp transmission maxima. In this experiment, it is used to determine the wavelength of a He–Ne laser and the wavelength separation of the sodium doublet. For the laser measurement, the number of fringes appearing or disappearing at the centre is correlated with the displacement of a movable mirror. For the sodium source, the recurrence of a clear fringe pattern is used to estimate the separation between the two nearby spectral lines. Using the adopted micrometer lever ratio $r = 0.10$, the mean laser wavelength estimate is 623.53 nm from 34 intervals. The sodium-doublet measurements give a combined mean separation of 0.5997 nm. Both results are discussed in relation to reference values, measurement resolution, and calibration assumptions.

Contents

Theory **1**

1 Part I: He–Ne Laser Wavelength **7**

1.1 Aim . . . 7
1.2 Experimental Procedure . . . 7
1.3 Theory and Working Formula . . . 7
1.4 Observations and Calculations . . . 8
1.4.1 Worked calculations . . . 9
1.5 Propagation of Measurement Uncertainty . . . 9
1.5.1 Reading and fringe-count uncertainties . . . 9
1.5.2 General propagation formula . . . 10
1.5.3 Worked uncertainty calculations . . . 10
1.5.4 Uncertainty of the reported mean . . . 10
1.6 Data Analysis . . . 11
1.7 Residual Analysis . . . 12
1.8 Discussion . . . 12
1.9 Sources of Error . . . 12
1.10 Conclusion . . . 13

2 Part II: Sodium Doublet Separation **14**

2.1 Aim . . . 14
2.2 Measurement model . . . 14
2.3 Observations and Calculations . . . 15
2.3.1 Sample calculation . . . 16
2.3.2 Coherence-length estimate . . . 16
2.4 Uncertainty and Statistical Analysis . . . 16
2.5 Comparison with the Theoretical Value . . . 18
2.6 Discussion . . . 19
2.7 Sources of Error . . . 19
2.8 Conclusion . . . 20

Limitations and Improvements **21**

Overall Conclusion **22**

References **23**

Theory

Principle of the Fabry–Pérot Interferometer

A Fabry–Pérot interferometer is an optical instrument that uses multiple-beam interference to measure wavelengths, resolve closely spaced spectral lines, and investigate the spectral properties of light. It consists of two highly reflecting surfaces aligned approximately parallel to one another, forming an optical cavity of separation d. Light entering the cavity undergoes repeated reflections, and the transmitted beams interfere with one another. For a ray propagating inside the cavity at an angle θ with respect to the normal to the reflecting surfaces, the optical path difference between successive transmitted beams is

$$\Delta = 2nd\cos\theta, \tag{1}$$

where n is the refractive index of the medium between the reflectors. For an air-filled cavity, $n \approx 1$. Constructive interference in transmission occurs when the optical path difference is an integral multiple of the wavelength:

$$2nd\cos\theta = m\lambda, \tag{2}$$

where m is the integer interference order and λ is the wavelength in the medium. When the wavelength is specified as the vacuum wavelength, the refractive index appears explicitly in the optical path difference. For a fixed cavity separation, rays satisfying the same interference condition emerge at the same inclination. Under appropriate illumination and observation conditions, the resulting fringes form concentric circular rings. At the centre of the pattern, $\theta \approx 0$, and the condition for constructive interference reduces to

$$2nd = m\lambda. \tag{3}$$

The fringe order therefore depends on the cavity separation, refractive index, and wavelength. A change in mirror separation changes the interference condition and causes fringes to appear, disappear, or shift across the observation field.

Interference Intensity

The interference pattern arises from the superposition of coherent optical fields. If two electric fields at the observation point are represented by complex amplitudes E_1 and E_2, the resultant field is

$$E = E_1 + E_2. \tag{4}$$

The time-averaged intensity is proportional to the squared magnitude of the resultant field:

$$I \propto |E_1 + E_2|^2 \tag{5}$$

$$= |E_1|^2 + |E_2|^2 + 2\,\mathrm{Re}(E_1 E_2^*). \tag{6}$$

For two beams with intensities I_1 and I_2 and a phase difference ϕ, the intensity becomes

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi. \tag{7}$$

Constructive interference occurs when the phase difference is an integer multiple of 2π, whereas destructive interference occurs when it is an odd multiple of π. In a Fabry–Pérot interferometer, several transmitted beams contribute to the observed intensity. Their phase differences depend on the optical path difference between successive beams. The resulting transmission pattern is therefore strongly dependent on the cavity separation and the wavelength of the incident radiation.

Micrometer Calibration and Mirror Displacement

The separation of the reflecting surfaces is adjusted using a micrometer mechanism. Let R_1 and R_2 denote two micrometer readings and let r be the calibration factor relating the change in micrometer reading to the actual mirror displacement. The magnitude of the mirror displacement is

$$|\Delta d| = r|R_2 - R_1|. \tag{8}$$

For the He–Ne laser calculation in Part I, the factor $r = 0.1$ is used as the assumed calibration factor for this analysis. This factor must be interpreted according to the units and calibration convention of the micrometer readings. It is not a universal property of the Fabry–Pérot interferometer. The displacement is calculated from the difference between the initial and final readings. The absolute value is used when determining the wavelength because the wavelength depends on the magnitude of the displacement rather than its direction. The lever-ratio conversion is specific to Part I. It is not used in the sodium doublet calculation in Part II, where the measured quantity is the cavity-separation interval associated with the recurrence of the fringe pattern.

Part I: Determination of the He–Ne Laser Wavelength

The wavelength of the He–Ne laser is determined by measuring the mirror displacement required for a known number of fringes to pass through a fixed observation point. At

normal incidence, the resonance condition is

$$2nd = m\lambda. \tag{9}$$

If the cavity separation changes by Δd, the optical path difference changes by $2n\Delta d$ at normal incidence. For a fixed wavelength, the corresponding change in interference order is therefore

$$N = \frac{2n|\Delta d|}{\lambda}, \tag{10}$$

where N is the number of complete fringe transitions observed. Rearranging gives the wavelength:

$$\boxed{\lambda = \frac{2n|\Delta d|}{N}}. \tag{11}$$

For an air-filled cavity, $n \approx 1$, so the expression reduces to

$$\boxed{\lambda = \frac{2|\Delta d|}{N}}. \tag{12}$$

Using the micrometer calibration in Equation (8), the wavelength can also be written as

$$\boxed{\lambda = \frac{2r|R_2 - R_1|}{N}}. \tag{13}$$

This relation is used exclusively for the He–Ne laser measurements in Part I. The measured wavelength is compared with the accepted reference value

$$\lambda_{\mathrm{He-Ne,lit}} = 632.8\,\mathrm{nm}. \tag{14}$$

The percentage deviation from the reference value is calculated using

$$\%\ \mathrm{Error} = \left|\frac{\lambda_{\mathrm{exp}} - \lambda_{\mathrm{lit}}}{\lambda_{\mathrm{lit}}}\right| \times 100. \tag{15}$$

Part II: Determination of Sodium Doublet Separation

Sodium emission contains two closely spaced yellow spectral lines, conventionally called the D_1 and D_2 lines. Their air wavelengths are approximately 589.592 nm and 588.995 nm, respectively [3]. Because the two wavelengths are close to one another, their interference patterns can overlap and produce a variation in the visibility of the resultant fringes. The visibility of an interference pattern is defined by

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, \tag{16}$$

where $I_{\max}$ and $I_{\min}$ are the maximum and minimum intensities, respectively. High visibility corresponds to clearly defined fringes, whereas low visibility corresponds to reduced fringe contrast. For two wavelengths λ_1 and λ_2, the resonance orders are slightly different at a given cavity separation. As the separation is varied, the two fringe systems periodically move into and out of coincidence. This produces a recurrence of high-visibility and low-visibility conditions. For near-normal incidence, the resonance conditions for the two wavelengths can be written as

$$2nd_1 = m\lambda_1, \qquad 2nd_2 = m\lambda_2, \tag{17}$$

for an appropriate pair of neighbouring resonance conditions of the two spectral components. The small wavelength difference is related to the cavity-separation interval associated with the recurrence of the interference pattern. For two closely spaced wavelengths with mean wavelength λ and separation $\Delta\lambda$, the relation used in this experiment is

$$\boxed{\Delta\lambda = \frac{\lambda^2}{2d_1}}. \tag{18}$$

Here, d_1 denotes the relevant measured cavity-separation interval between successive recurrence conditions, according to the experimental measurement convention. It must not be confused with the absolute cavity separation unless that is how the apparatus readings define it. For the sodium doublet, the mean wavelength is taken as

$$\lambda \approx 589.3\,\text{nm}. \tag{19}$$

If d_1 is measured in millimetres, it must be converted to nanometres before substitution into Equation (18). Since $1\,\text{mm} = 10^6\,\text{nm}$, the doublet separation can be expressed as

$$\boxed{\Delta\lambda(\text{nm}) = \frac{(589.3)^2}{2d_1(\text{mm}) \times 10^6}}. \tag{20}$$

The experimentally determined separation is compared with the accepted difference between the sodium D lines:

$$\Delta\lambda_{\text{lit}} = |\lambda_{D_1} - \lambda_{D_2}| \tag{21}$$
$$= |589.592 - 588.995|\,\text{nm} \tag{22}$$
$$= 0.597\,\text{nm}. \tag{23}$$

The percentage error for Part II is therefore

$$\%\ \text{Error}_{\text{Na}} = \left|\frac{\Delta\lambda_{\text{exp}} - \Delta\lambda_{\text{lit}}}{\Delta\lambda_{\text{lit}}}\right| \times 100. \tag{24}$$

No lever-ratio correction is applied to this calculation unless the experimental calibration specifically requires one. The measured interval must be used in the units and convention defined by the observation table.

Uncertainty Propagation

The uncertainty in the calculated quantities depends on the precision of the micrometer readings, the uncertainty in the fringe count, and the repeatability of the observed fringe pattern. For the wavelength determination in Part I,

$$\lambda = \frac{2r|\Delta R|}{N}. \tag{25}$$

Assuming independent uncertainties in r, ΔR, and N, the relative uncertainty is

$$\left(\frac{\delta\lambda}{\lambda}\right)^2 = \left(\frac{\delta r}{r}\right)^2 + \left(\frac{\delta(\Delta R)}{|\Delta R|}\right)^2 + \left(\frac{\delta N}{N}\right)^2. \tag{26}$$

If the calibration factor is treated as exact for the calculation, the first term is omitted. The absolute uncertainty is then

$$\delta\lambda = \lambda\frac{\delta\lambda}{\lambda}. \tag{27}$$

For Part II, the doublet separation is

$$\Delta\lambda = \frac{\lambda^2}{2d_1}. \tag{28}$$

If the uncertainties in λ and d_1 are independent, the propagated relative uncertainty is

$$\left(\frac{\delta(\Delta\lambda)}{\Delta\lambda}\right)^2 = \left(2\frac{\delta\lambda}{\lambda}\right)^2 + \left(\frac{\delta d_1}{d_1}\right)^2. \tag{29}$$

If the mean sodium wavelength is treated as fixed at 589.3 nm, the first term is omitted, giving

$$\boxed{\delta(\Delta\lambda) = \Delta\lambda\frac{\delta d_1}{d_1}}. \tag{30}$$

The statistical uncertainty of repeated measurements is assessed separately using the sample standard deviation and the standard error of the mean. The standard error is

$$\mathrm{SEM} = \frac{s}{\sqrt{n}}, \tag{31}$$

where s is the sample standard deviation and n is the number of valid measurements. Statistical uncertainty and propagated instrumental uncertainty represent different contributions and should be reported separately.

Scope of the Theory

The theory is divided according to the physical quantity determined in each part of the experiment.

- **Common theory:** Optical path difference, constructive interference, and the formation of interference fringes provide the foundation for both parts.
- **Part I – He–Ne laser:** The micrometer calibration, mirror displacement, fringe count, wavelength equation, and associated uncertainty propagation are used to determine the laser wavelength.
- **Part II – Sodium doublet:** The recurrence of fringe visibility, the mean sodium wavelength, the measured cavity-separation interval, and the doublet separation equation are used to estimate the wavelength difference between the sodium D lines.
- **Separate calibration conventions:** The assumed lever ratio $r = 0.1$ is used only for Part I. Part II uses the recorded separation interval directly, without applying that lever ratio.

Part I: He–Ne Laser Wavelength

1.1 Aim

To determine the wavelength of a He–Ne laser using a Fabry–Pérot interferometer from the measured mirror displacement corresponding to a known number of fringes passing through the centre of the interference pattern.

1.2 Experimental Procedure

The Fabry–Pérot interferometer was adjusted until a clear circular fringe pattern was obtained. The initial micrometer reading was recorded, and the number of fringes passing through the centre was counted as the movable mirror was displaced. The total reading was recorded after each interval. The change between consecutive readings was converted to mirror displacement using the single calibration factor applied throughout the supplied workbook, $r = 0.10$. The wavelength was then calculated for each interval.

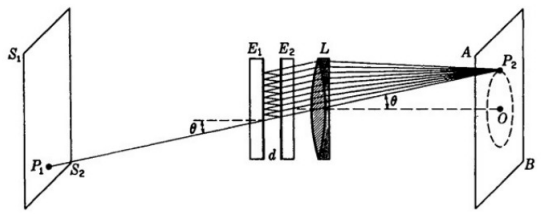


Figure 1.1: Experimental setup of the Fabry–Pérot interferometer.



Figure 1.2: Circular fringe pattern observed during the experiment.

1.3 Theory and Working Formula

For a Fabry–Pérot interferometer, constructive transmission at near-normal incidence occurs when

$$2d = m\lambda, \tag{1.1}$$

where d is the mirror separation, m is the interference order, and λ is the wavelength. When the mirror separation changes by x, the optical path difference changes by $2x$. If N fringes pass a fixed reference point,

$$2x = N\lambda. \tag{1.2}$$

Therefore,

$$\boxed{\lambda = \frac{2x}{N}}. \tag{1.3}$$

For the micrometer readings used here, the actual mirror displacement is $x = r|\Delta R|$, with $r = 0.10$. Thus, when ΔR is in millimetres, the wavelength in nanometres is

$$\boxed{\lambda\,(\mathrm{nm}) = \frac{2r|\Delta R|}{N} \times 10^{6}}. \tag{1.4}$$

The calibration factor is retained as adopted in the supplied workbook; its physical value should be confirmed from the instrument specification.

1.4 Observations and Calculations

The initial total reading was $R_1 = 9.50\,\mathrm{mm}$. For each subsequent observation, the change in reading and the corresponding mirror displacement were calculated as

$$|\Delta R_i| = |R_i - R_{i-1}|, \tag{1.5}$$

$$\Delta d_i = r|\Delta R_i| = 0.10|\Delta R_i|. \tag{1.6}$$

The wavelength estimate for an interval containing N_i fringes is

$$\lambda_i = \frac{2\Delta d_i}{N_i} \times 10^{6}\,\mathrm{nm}, \tag{1.7}$$

where the displacement is in millimetres. The same calibration factor, $r = 0.10$, is used in every row below.

Table 1.1: Recorded readings and calculated wavelength estimates ($r = 0.10$). The table is split into two parallel panels to use the page width efficiently.

No.	N	R (mm)	$\lvert\Delta R\rvert$ (mm)	Δd (mm)	λ (nm)
1	–	9.50	–	–	–
2	10	9.47	0.03	0.003	600
3	10	9.44	0.03	0.003	600
4	10	9.40	0.04	0.004	800
5	10	9.37	0.03	0.003	600
6	10	9.34	0.03	0.003	600
7	10	9.31	0.03	0.003	600
8	10	9.27	0.04	0.004	800
9	10	9.24	0.03	0.003	600
10	10	9.21	0.03	0.003	600
11	10	9.18	0.03	0.003	600
12	10	9.15	0.03	0.003	600
13	10	9.12	0.03	0.003	600
14	10	9.09	0.03	0.003	600
15	10	9.06	0.03	0.003	600
16	10	9.03	0.03	0.003	600
17	10	8.99	0.04	0.004	800
18	10	8.95	0.04	0.004	800
19	10	8.92	0.03	0.003	600
20	10	8.89	0.03	0.003	600
21	10	8.86	0.03	0.003	600
22	10	8.83	0.03	0.003	600
23	10	8.80	0.03	0.003	600
24	20	8.74	0.06	0.006	600
25	20	8.68	0.06	0.006	600
26	20	8.62	0.06	0.006	600
27	20	8.56	0.06	0.006	600
28	20	8.50	0.06	0.006	600
29	20	8.43	0.07	0.007	700
30	20	8.37	0.06	0.006	600
31	20	8.31	0.06	0.006	600
32	20	8.25	0.06	0.006	600
33	20	8.19	0.06	0.006	600
34	20	8.14	0.05	0.005	500
35	30	8.05	0.09	0.009	600

1.4.1 Worked calculations

For the first interval, the readings are $R_1 = 9.50\,\text{mm}$ and $R_2 = 9.47\,\text{mm}$, with $N = 10$ fringes. Therefore,

$$|\Delta R| = |9.47 - 9.50| = 0.03\,\text{mm}, \tag{1.8}$$

$$\Delta d = 0.10(0.03) = 0.003\,\text{mm}, \tag{1.9}$$

$$\lambda = \frac{2(0.003)}{10} \times 10^6 = 600\,\text{nm}. \tag{1.10}$$

For an interval with $|\Delta R| = 0.04\,\text{mm}$ and $N = 10$,

$$\lambda = \frac{2[0.10(0.04)]}{10} \times 10^6 = 800\,\text{nm}. \tag{1.11}$$

For the interval with $|\Delta R| = 0.07\,\text{mm}$ and $N = 20$,

$$\lambda = \frac{2[0.10(0.07)]}{20} \times 10^6 = 700\,\text{nm}. \tag{1.12}$$

For the interval with $|\Delta R| = 0.05\,\text{mm}$ and $N = 20$,

$$\lambda = \frac{2[0.10(0.05)]}{20} \times 10^6 = 500\,\text{nm}. \tag{1.13}$$

1.5 Propagation of Measurement Uncertainty

1.5.1 Reading and fringe-count uncertainties

For an interval containing N_i counted fringes, the measured wavelength is

$$\lambda_i = \frac{2r|R_i - R_{i-1}|}{N_i} \times 10^6\,\text{nm}, \qquad r = 0.10. \tag{1.14}$$

The micrometer readings are reported in steps of $q = 0.01\,\text{mm}$. If the readings were rounded to the nearest division and rounding error is uniformly distributed, the standard uncertainty of one reading is

$$u_R = \frac{q}{\sqrt{12}} = 0.002887\,\text{mm}. \tag{1.15}$$

Assuming the two endpoint rounding errors are independent, the standard uncertainty of their difference is

$$u_{\Delta R} = \sqrt{u_{R_i}^2 + u_{R_{i-1}}^2} = \sqrt{2}\,u_R = \frac{0.01}{\sqrt{6}} = 0.004082\,\text{mm}. \tag{1.16}$$

To allow for counting ambiguity, a **conservative assumed** standard uncertainty of $u_N = 1$ fringe is used. This value is not measured from the data; it should be replaced if the experimental counting uncertainty is known. The calibration factor is used consistently as $r = 0.10$. Its standard uncertainty u_r was not supplied, so its contribution is retained

symbolically and omitted only in the numerical examples below.

1.5.2 General propagation formula

For independent uncertainty contributions, first-order propagation gives

$$u_{\lambda_i}^2 = \left(\frac{\partial \lambda_i}{\partial r} u_r\right)^2 + \left(\frac{\partial \lambda_i}{\partial |\Delta R_i|} u_{\Delta R}\right)^2 + \left(\frac{\partial \lambda_i}{\partial N_i} u_N\right)^2, \tag{1.17}$$

$$\frac{u_{\lambda_i}}{\lambda_i} = \sqrt{\left(\frac{u_r}{r}\right)^2 + \left(\frac{u_{\Delta R}}{|\Delta R_i|}\right)^2 + \left(\frac{u_N}{N_i}\right)^2}. \tag{1.18}$$

Thus,

$$\boxed{u_{\lambda_i} = \lambda_i \sqrt{\left(\frac{u_r}{0.10}\right)^2 + \left(\frac{0.004082}{|\Delta R_i|}\right)^2 + \left(\frac{1}{N_i}\right)^2}} \tag{1.19}$$

where $|\Delta R_i|$ is expressed in millimetres. This formula assumes small, approximately independent input uncertainties; it is an estimate, particularly for intervals with only a few micrometer divisions of travel.

1.5.3 Worked uncertainty calculations

For observation 2, $|\Delta R| = 0.03\,\text{mm}$, $N = 10$, and $\lambda = 600\,\text{nm}$. Neglecting the unspecified calibration uncertainty,

$$u_{\lambda,R} = 600 \left(\frac{0.004082}{0.03}\right) = 81.65\,\text{nm}, \tag{1.20}$$

$$u_{\lambda,N} = 600 \left(\frac{1}{10}\right) = 60.00\,\text{nm}, \tag{1.21}$$

$$u_\lambda = \sqrt{81.65^2 + 60.00^2} = 101.32\,\text{nm}. \tag{1.22}$$

The illustrative interval result is therefore

$$\lambda_2 = (600 \pm 101)\,\text{nm} \qquad \text{(standard uncertainty, assuming } u_N = 1\text{)}. \tag{1.23}$$

For observation 29, $|\Delta R| = 0.07\,\text{mm}$, $N = 20$, and $\lambda = 700\,\text{nm}$:

$$u_{\lambda,29} = 700 \sqrt{\left(\frac{0.004082}{0.07}\right)^2 + \left(\frac{1}{20}\right)^2} = 53.77\,\text{nm}. \tag{1.24}$$

The longer counting interval has a smaller relative uncertainty.

1.5.4 Uncertainty of the reported mean

The sample standard deviation and standard error in the following section characterise the observed scatter of the 34 interval estimates. They are not a substitute for the instrument and fringe-count uncertainty budget. Moreover, consecutive intervals share endpoint readings, so their reading errors are correlated; any calibration-factor uncer-

tainty is common to all intervals. Consequently, the individual propagated uncertainties should *not* simply be divided by $\sqrt{34}$ to claim a complete uncertainty on the mean. The result below reports the empirical standard error separately, with calibration uncertainty unquantified.

1.6 Data Analysis

A total of $n = 34$ wavelength estimates were obtained. The values consist of 28 estimates of 600 nm, four of 800 nm, one of 700 nm, and one of 500 nm. The arithmetic mean is

$$\bar{\lambda} = \frac{1}{n}\sum_{i=1}^{n}\lambda_i \tag{1.25}$$

$$= \frac{28(600) + 4(800) + 700 + 500}{34}\,\text{nm} \tag{1.26}$$

$$= 623.53\,\text{nm}. \tag{1.27}$$

The sample standard deviation is

$$s = \sqrt{\frac{\sum_{i=1}^{n}(\lambda_i - \bar{\lambda})^2}{n-1}} = 69.89\,\text{nm}, \tag{1.28}$$

and the standard error of the mean is

$$\text{SEM} = \frac{s}{\sqrt{n}} = \frac{69.89}{\sqrt{34}} = 11.99\,\text{nm}. \tag{1.29}$$

Thus, the statistical summary is

$$\boxed{\bar{\lambda} = (623.53 \pm 11.99)\,\text{nm}}, \tag{1.30}$$

where the quoted uncertainty is the standard error of the mean and does not include a full instrumental uncertainty budget. Using 632.8 nm as the nominal wavelength of a He–Ne laser, the absolute difference is

$$|\Delta\lambda| = |623.53 - 632.8| = 9.27\,\text{nm}, \tag{1.31}$$

$$\text{Percentage deviation} = \frac{|623.53 - 632.8|}{632.8} \times 100 \tag{1.32}$$

$$\approx 1.47\%. \tag{1.33}$$

1.7 Residual Analysis

For each observation, the residual was defined as $e_i = \lambda_i - \bar{\lambda}$, where $\bar{\lambda} = 623.53\,\text{nm}$. Figure 1.3 plots these residuals against observation number. Residuals above zero correspond to estimates greater than the sample mean, while negative residuals correspond to estimates below it. The discrete bands arise because the micrometer readings are recorded to 0.01 mm, so consecutive-reading differences take only a small set of values. The plot is therefore useful for visualising the quantisation and spread of the estimates, rather than evidence of a continuous random measurement distribution.

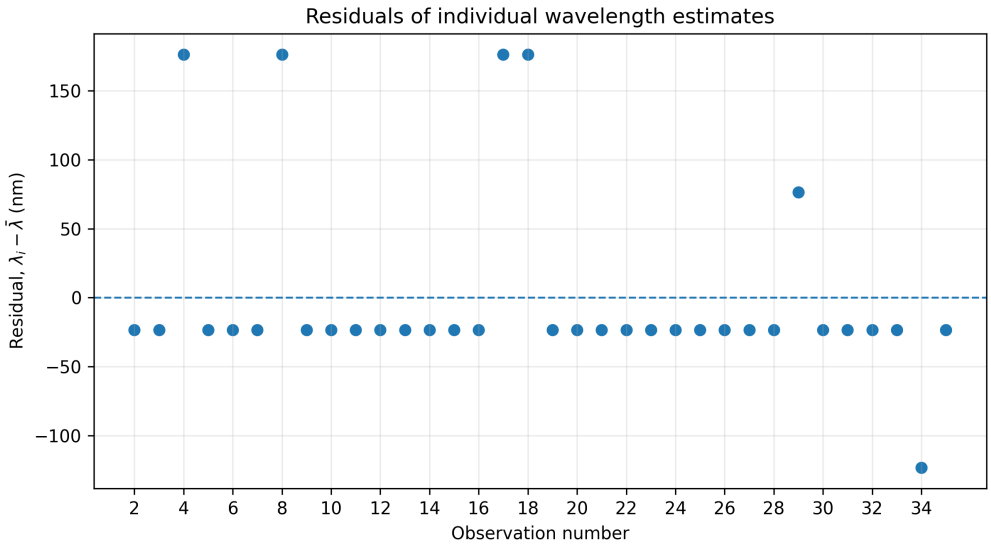


Figure 1.3: Residuals of the individual wavelength estimates relative to the sample mean, $e_i = \lambda_i - \bar{\lambda}$. The dashed horizontal line marks zero residual.

1.8 Discussion

With the workbook's calibration factor $r = 0.10$ applied consistently, the mean wavelength estimate is 623.53 nm, which is 1.47% below the nominal value of 632.8 nm. However, the individual estimates show substantial scatter ($s = 69.89\,\text{nm}$). This scatter is strongly affected by the limited precision of the recorded micrometer readings: changes of 0.03 and 0.04 mm, for example, lead to distinct wavelength estimates of 600 and 800 nm when $N = 10$. Consequently, the closeness of the mean to the nominal wavelength should be interpreted cautiously. The value $r = 0.10$ is consistent with the calculation formulas in the supplied workbook; its physical correctness depends on the instrument's actual calibration specification and cannot be established from the readings alone.

1.9 Sources of Error

1. **Micrometer resolution:** readings recorded to 0.01 mm make the difference between consecutive readings coarse, contributing strongly to the spread in calculated wavelengths.

2. **Calibration uncertainty:** any error in the conversion factor between micrometer reading and actual mirror displacement produces a systematic error in the wavelength.

3. **Fringe counting:** fringes may be missed or counted twice, particularly if the pattern has low contrast or moves rapidly.

4. **Mirror alignment:** imperfect alignment can distort the circular fringes and obscure the centre.

5. **Mechanical backlash and vibration:** backlash in the adjustment mechanism and environmental vibration can make the actual mirror motion differ from the indicated motion.

1.10 Conclusion

The He–Ne laser wavelength was estimated from 34 fringe-counting intervals using the Fabry–Pérot interferometer. Applying the single calibration factor $r = 0.10$ throughout gives a mean of 623.53 nm, a sample standard deviation of 69.89 nm, and a standard error of 11.99 nm. The mean differs from the nominal wavelength of 632.8 nm by 1.47%. The residual plot and the large spread of individual estimates indicate that the reading resolution limits the precision of this method as applied here. The calibration factor should be checked against the instrument specification, and longer measurement intervals spanning more fringes would help reduce the relative effect of reading resolution.

Part II: Sodium Doublet Separation

2.1 Aim

To determine the wavelength separation of the two closely spaced spectral lines of the sodium doublet using the recurrence of coincident fringe patterns in a Fabry–Pérot interferometer.

2.2 Measurement model

For a Fabry–Pérot cavity at near-normal incidence, a transmitted wavelength satisfies the resonance condition $2nd = m\lambda$, where d is the mirror separation, n is the refractive index inside the cavity, and m is an integer interference order. Sodium light contains two nearby wavelengths, λ_1 and λ_2. Their fringe systems therefore have slightly different phases: as the cavity separation changes, the fringes move relative to one another, alternating between clear coincidence and reduced visibility. The measured interval d_1 is the mirror travel between successive clear coincidences.

The recurrence relation follows from the change in relative phase over a cavity-separation change D. For a small wavelength difference $\Delta\lambda = |\lambda_1 - \lambda_2|$ about the mean wavelength λ, the relative phase changes approximately as

$$\Delta\phi \simeq 4\pi D \frac{\Delta\lambda}{\lambda^2}.$$

One full recurrence occurs when this phase changes by 2π, giving $D \simeq \lambda^2/(2\Delta\lambda)$. Taking $D = d_1$ yields Eq. (20). The calculation uses the adopted mean sodium wavelength $\lambda = 589.3\,\mathrm{nm}$ and treats the recorded interval as actual mirror travel, as specified by the supplied worksheet. Any lever ratio or scale conversion should be applied only if the instrument calibration requires it; this convention should be checked against the instrument specification.

The same two-line spectrum has an approximate coherence length

$$L_c \approx \frac{\lambda^2}{\Delta\lambda} \approx 2d_1. \tag{2.1}$$

Here L_c is the optical-path-difference scale over which the two spectral components remain mutually coherent; it is not the coherence length of either sodium line considered separately. The factor of two arises because moving one cavity mirror by d_1 changes the round-trip optical path by approximately $2d_1$. These relations assume $\Delta\lambda \ll \lambda$ and near-normal incidence, and the measured value remains sensitive to how consistently the clearest coincidence is identified.

2.3 Observations and Calculations

Two sets of ten measurements were supplied. MSR denotes the main-scale reading and CSR the circular-scale reading. The displacement d_1 and the corresponding wavelength separation were taken from the supplied worksheet. The wavelength separations shown below are rounded to three decimal places; the statistical analysis was performed using values calculated from the listed displacements before rounding.

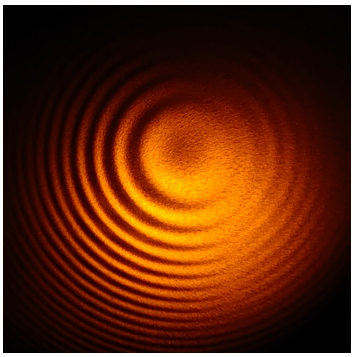


Figure 2.1: Sodium-doublet fringe pattern.

Table 2.1: Sodium-doublet observations and calculated wavelength separation.

Set 1				
Trial	MSR	CSR	d_1 (mm)	$\Delta\lambda$ (nm)
1	10.5	42	0.21	0.827
2	10.5	21	0.19	0.914
3	10.0	20	0.32	0.543
4	9.5	41	0.29	0.599
5	9.5	8	0.33	0.526
6	9.0	17	0.41	0.424
7	8.5	38	0.29	0.599
8	8.5	7	0.31	0.560
9	8.0	27	0.30	0.579
10	7.5	45	0.32	0.543

Set 2				
Trial	MSR	CSR	d_1 (mm)	$\Delta\lambda$ (nm)
1	10.0	22	0.21	0.827
2	9.5	47	0.25	0.695
3	9.5	14	0.33	0.526
4	9.0	25	0.39	0.445
5	8.5	46	0.29	0.599
6	8.5	12	0.34	0.511
7	8.0	32	0.32	0.543
8	8.0	2	0.30	0.579
9	7.5	21	0.31	0.560
10	7.0	42	0.29	0.599

The two sets each contain ten measurements, allowing the repeatability of the observed coincidence interval to be compared across trials. The model predicts an inverse relationship: a larger measured d_1 gives a smaller calculated wavelength separation $\Delta\lambda$. These individual estimates are used in the sample calculation below and in the subsequent statistical analysis.

2.3.1 Sample calculation

For Set 1, Trial 1, the measured displacement is $d_1 = 0.21\,\text{mm} = 0.21 \times 10^{-3}\,\text{m}$. Substituting $\lambda = 589.3\,\text{nm} = 589.3 \times 10^{-9}\,\text{m}$ into Eq. (18) gives

$$\Delta\lambda = \frac{(589.3 \times 10^{-9}\,\text{m})^2}{2(0.21 \times 10^{-3}\,\text{m})} \tag{2.2}$$

$$= 8.2684 \times 10^{-10}\,\text{m} \tag{2.3}$$

$$\approx 0.827\,\text{nm}. \tag{2.4}$$

The same calculation was applied to every displacement measurement.

2.3.2 Coherence-length estimate

For Set 1, Trial 1, $d_1 = 0.21\,\text{mm}$. Applying Eq. (2.1) gives

$$L_c \approx 2(0.21\,\text{mm}) = 0.42\,\text{mm}. \tag{2.5}$$

This is the coherence length associated with the two sodium spectral components, not that of either individual line.

2.4 Uncertainty and Statistical Analysis

The readings were recorded to a resolution of 0.01 mm. Assuming that the rounding error is uniformly distributed over an interval of width 0.01 mm, the standard uncertainty of one reading is

$$u_R = \frac{0.01}{\sqrt{12}} = 0.00289\,\text{mm}. \tag{2.6}$$

Since d_1 is obtained from the difference between two independent readings, the resolution-related uncertainty in the displacement is

$$u_{d_1,\text{res}} = \sqrt{u_R^2 + u_R^2} = \sqrt{2}\,u_R = 0.00408\,\text{mm}. \tag{2.7}$$

Other possible contributions include uncertainty in locating the clearest fringe coincidence, micrometer backlash, vibration, and displacement calibration. If these contributions are independent, the total displacement uncertainty is

$$u_{d_1} = \sqrt{u_{d_1,\text{res}}^2 + u_{d_1,\text{pattern}}^2 + u_{d_1,\text{backlash}}^2 + u_{d_1,\text{cal}}^2}. \tag{2.8}$$

The available worksheet does not quantify these additional terms, so the complete instrumental uncertainty cannot be determined. For $\Delta\lambda = \lambda^2/(2d_1)$, propagation of independent uncertainties gives

$$u_{\Delta\lambda} = \Delta\lambda \sqrt{\left(\frac{2u_\lambda}{\lambda}\right)^2 + \left(\frac{u_{d_1}}{d_1}\right)^2}. \tag{2.9}$$

The uncertainty in the adopted mean wavelength is not specified. Treating λ as fixed and including only reading resolution gives

$$u_{\Delta\lambda_i,\mathrm{res}} = \Delta\lambda_i \frac{0.00408\,\mathrm{mm}}{d_{1,i}}, \tag{2.10}$$

where $d_{1,i}$ is in millimetres. This is a partial standard-uncertainty estimate, not a maximum error or a complete uncertainty budget. Since the working formula does not explicitly use a counted number of fringes, uncertainty in identifying successive clear patterns is included in the displacement uncertainty.

For each dataset, the mean, sample standard deviation, and standard error of the mean (SEM) were calculated from the ten unrounded wavelength-separation estimates:

$$\overline{\Delta\lambda} = \frac{1}{n}\sum_{i=1}^{n}\Delta\lambda_i, \tag{2.11}$$

$$s = \sqrt{\frac{\sum_{i=1}^{n}(\Delta\lambda_i - \overline{\Delta\lambda})^2}{n-1}}, \tag{2.12}$$

$$u_{\mathrm{stat}} = \frac{s}{\sqrt{n}}. \tag{2.13}$$

The standard deviation describes the spread of individual readings, while the SEM estimates the statistical uncertainty in the mean. Neither includes systematic errors by itself.

Since Eq. (2.1) gives $L_{c,i} \approx 2d_{1,i}$ for each trial, the coherence-length statistics are summarised in Table 2.3.

Table 2.2: Wavelength-separation statistics (nm).

Dataset	Mean	SD	SEM
Set 1 ($n = 10$)	0.6112	0.1468	0.0464
Set 2 ($n = 10$)	0.5883	0.1063	0.0336
Combined ($n = 20$)	0.5997	0.1253	0.0280

Table 2.3: Coherence-length statistics (mm).

Dataset	Mean	SD	SEM
Set 1 ($n = 10$)	0.594	0.123	0.039
Set 2 ($n = 10$)	0.606	0.098	0.031
Combined ($n = 20$)	0.600	0.109	0.024

For the combined measurements, the mean coherence length is 0.600 mm, with a statistical standard error of approximately 0.024 mm. The combined mean wavelength separation is 0.5997 nm, with an SEM of 0.0280 nm. The estimated resolution-related contribution to the uncertainty in the combined mean is approximately 0.0021 nm. Combining these contributions in quadrature gives

$$u_{\mathrm{combined}} = \sqrt{(0.0280)^2 + (0.0021)^2}\,\mathrm{nm} \approx 0.0281\,\mathrm{nm}. \tag{2.14}$$

The experimental result, quoted with one-standard-uncertainty precision, is therefore

$$\boxed{\Delta\lambda = (0.600 \pm 0.028)\,\mathrm{nm}}. \tag{2.15}$$

This estimate includes the statistical spread and the estimated reading-resolution contribution, but excludes systematic contributions that have not been quantified.

The residual for each measurement is the difference between its wavelength-separation estimate and the mean of its own dataset:

$$r_i = \Delta\lambda_i - \overline{\Delta\lambda}. \tag{2.16}$$

Residuals scattered around zero without a clear trend are consistent with random variation. A trend or unusually large residuals may point to changes in alignment, inconsistent identification of fringe coincidence, or an anomalous displacement reading. The residual plot should be interpreted with the limited number of measurements in mind; it cannot establish that systematic errors are absent.

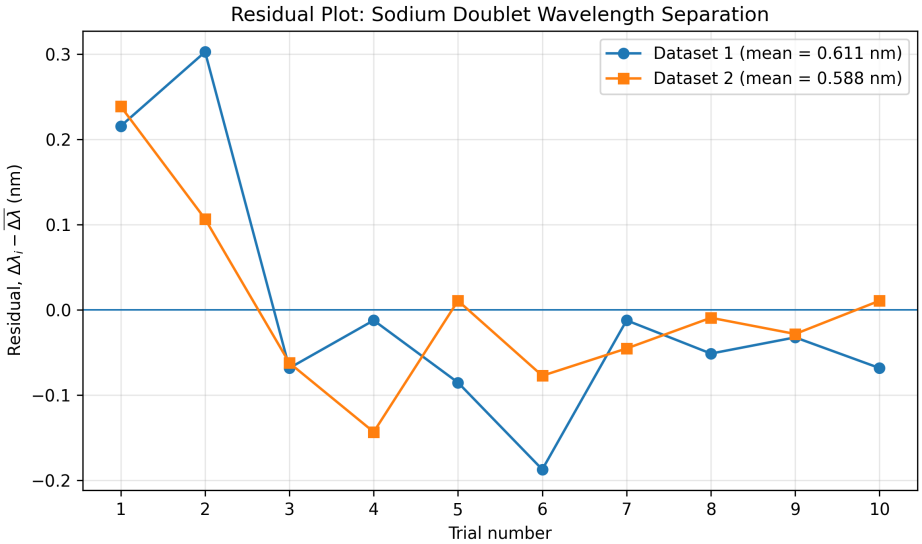


Figure 2.2: Residuals of the sodium-doublet wavelength-separation estimates for both datasets, calculated relative to each dataset's mean.

2.5 Comparison with the Theoretical Value

The accepted air wavelengths of the sodium D lines are approximately 588.9950 nm and 589.5924 nm [3]. Their separation is therefore

$$\Delta\lambda_{\text{th}} = 589.5924 - 588.9950 = 0.5974\,\text{nm}. \tag{2.17}$$

Comparing this value with the combined experimental mean, $\Delta\lambda_{\text{exp}} = 0.5997\,\text{nm}$, the absolute difference is

$$|\Delta\lambda_{\text{exp}} - \Delta\lambda_{\text{th}}| = |0.5997 - 0.5974| = 0.0023\,\text{nm}. \tag{2.18}$$

The percentage deviation from the theoretical value is

$$\%\ \text{deviation} = \frac{|\Delta\lambda_{\text{exp}} - \Delta\lambda_{\text{th}}|}{\Delta\lambda_{\text{th}}} \times 100 = \frac{0.0023}{0.5974} \times 100 \approx 0.39\%. \tag{2.19}$$

The experimental mean is about 0.39% higher than the theoretical separation. This difference is small compared with the reported SEM of 0.0280 nm, so the experimental

Table 2.4: Comparison of the experimental and theoretical sodium-doublet separation.

Quantity	Value (nm)
Theoretical separation	0.5974
Experimental mean separation	0.5997
Absolute difference	0.0023
Percentage deviation	0.39%

result is consistent with the theoretical value within the stated statistical uncertainty. This comparison does not account for unquantified systematic errors.

2.6 Discussion

The two measurement sets yield mean wavelength separations of 0.6112 nm and 0.5883 nm; their combined mean is 0.5997 nm. The accepted air wavelengths of the sodium D lines are approximately 588.9950 nm and 589.5924 nm [3], giving a reference separation of 0.5974 nm. The measured mean differs by 0.39%, which is small relative to the reported SEM of 0.0280 nm. This indicates agreement within the statistical uncertainty, although the calibration assumption and unquantified systematic contributions remain important. Using the individual displacement measurements, the mean estimated coherence length associated with the sodium doublet is 0.600 mm (SEM 0.024 mm). This value follows from $L_c \approx 2d_1$ for each trial. It is a spectral-separation estimate and not a direct measurement of the coherence length of either individual sodium D-line. The relation in Eq. (18) shows that $\Delta\lambda$ is inversely proportional to d_1. Consequently, an overestimate of the displacement leads to an underestimate of the inferred wavelength separation, and vice versa. The scatter among trials is consistent with the sensitivity of the calculation to small changes in measured displacement and with the difficulty of identifying the point of clearest fringe coincidence. The supplied worksheet treats the listed displacements as actual mirror displacements and applies no lever-ratio correction. This convention has been retained to preserve consistency with the provided data, but it must be verified against the instrument manual before the result is treated as an absolute measurement. The uncertainty quoted in Eq. (2.15) is primarily statistical; calibration, backlash, and pattern-identification uncertainties could increase the total uncertainty.

2.7 Sources of Error

1. **Reading resolution:** The scale readings are recorded to 0.01 mm, limiting the precision of the displacement difference.

2. **Fringe-coincidence identification:** The position of maximum clarity is judged visually, so the selected position may vary between trials.

3. **Displacement calibration:** A mismatch between the micrometer reading and actual mirror movement produces a systematic error. The no-lever-ratio assumption in the worksheet should be checked against the apparatus specification.

4. **Mechanical backlash:** Reversing the micrometer direction or adjusting the screw can introduce hysteresis into the displacement measurement.

5. **Optical alignment:** Non-parallel reflecting surfaces or imperfect alignment can distort the fringe pattern and make coincidence harder to identify.

6. **Vibration and air currents:** Mechanical vibration and fluctuations in the surrounding air can reduce fringe stability and visibility.

7. **Reference wavelength:** The adopted mean wavelength of 589.3 nm is treated as fixed in the uncertainty estimate; its uncertainty is not included.

2.8 Conclusion

The wavelength separation of the sodium doublet was estimated using the recurrence of clear fringe coincidences in a Fabry–Pérot interferometer. The two datasets yielded mean values of 0.6112 nm and 0.5883 nm, and the combined mean was 0.5997 nm. Including the statistical uncertainty and estimated reading-resolution contribution gives

$$\Delta\lambda = (0.600 \pm 0.028)\,\text{nm}.$$

The result differs by approximately 0.39% from the reference sodium D-line separation of about 0.5974 nm. The coherence length estimated from the individual displacement measurements is $L_c \approx (0.600 \pm 0.024)$ mm, where the uncertainty is the SEM and does not include unquantified systematic effects. The quoted uncertainty is not a complete uncertainty budget because systematic effects associated with displacement calibration, backlash, and fringe-position identification have not been quantified. In particular, the assumption that no lever-ratio correction is required should be checked against the instrument specification.

Limitations and Improvements

The results are consistent with the reference values, but several assumptions limit how strongly that agreement can be interpreted:

1. **Displacement calibration:** Part I uses $r = 0.10$, whereas Part II treats the tabulated interval as actual mirror travel. Verify both conventions against the interferometer manual or an independent length standard before reporting absolute wavelengths.

2. **Resolution and dependence:** In Part I, neighbouring interval estimates share micrometer readings, so the usual SEM calculation may not describe independent observations. Use longer fringe-count intervals and assess uncertainty from the raw readings or independent runs.

3. **Visual coincidence criterion:** The sodium recurrence positions are selected by eye. Define a consistent criterion for a clear coincidence, approach each setting from the same direction, and repeat the full run to assess repeatability and backlash.

4. **Incomplete systematic budget:** The quoted uncertainties do not quantify calibration, backlash, and pattern-identification contributions. Measure or bound these terms and combine them with the statistical uncertainty before making a precision claim.

Overall Conclusion

The experiment demonstrated two applications of multiple-beam interference in precision optical measurements using a Fabry–Pérot interferometer. In Part I, the wavelength of the He–Ne laser was determined from the relationship between mirror displacement and fringe-order change. Using the adopted lever ratio of $r = 0.10$, the mean wavelength obtained from 34 interval estimates was 623.53 nm. Compared with the nominal wavelength of 632.8 nm, the result has a relative deviation of approximately 1.47%. This value remains subject to verification of the lever ratio and the consistency of the recorded readings. In Part II, the separation of the sodium doublet was determined from the mirror displacement between successive clear fringe patterns. The two datasets yielded mean separations of 0.6112 nm and 0.5883 nm, giving a combined mean of 0.5997 nm. Compared with the reference value of approximately 0.5974 nm, the experimental result has a relative deviation of about 0.39%. The calculation follows the supplied workbook, in which no lever-ratio correction is applied to the sodium-doublet displacement. Overall, the results illustrate the use of fringe displacement and fringe-pattern recurrence to determine optical wavelengths and small wavelength differences. The accuracy of both measurements depends on displacement calibration, mirror alignment, mechanical stability, and reliable identification of the interference pattern. Verification of the instrument calibration and a more complete assessment of measurement uncertainties would improve the reliability of the reported results.

References

1. Supplier manual, *Fabry–Pérot Interferometer*, NISER laboratory handout, last updated September 2023.

2. F. A. Jenkins and H. E. White, *Fundamentals of Optics*.

3. A. Kramida, Yu. Ralchenko, J. Reader, and NIST ASD Team, *NIST Atomic Spectra Database*, version 5.12, Sodium I line wavelengths, 2024, accessed 23 October 2026, https://physics.nist.gov/ASD.